

## **Practical 2: Security Requirement Engineering with SQUARE Methodology**

**SQUARE** represents **Security Quality Requirements Engineering**. It is a requirements engineering method created by Carnegie Mellon University's Software Engineering Institute (SEI) that focuses on gathering and documenting security needs.

### **Source:**

<https://www.modernanalyst.com/Careers/InterviewQuestions/tabid/128/ID/2105/Describe-the-requirement-engineering-process-SQUARE.aspx>

### **Scenario:**

You are a member of a team that is creating a safe **online share trading** system. You must use the **SQUARE** model to identify and prioritize essential security requirements. Work in a team of 4-5 persons, perform the following steps (the steps listed are part of the steps in **SQUARE**).

#### **1. Define Security Goals (Step 2 – Identify security goals)**

- Define the **high-level security goals** for the system.

#### **To Do:**

What should the system protect? (e.g., data integrity, user privacy, transaction authenticity)

#### **2. Use Cases (Step 3 - Develop artifact)**

#### **To Do:**

Describe **THREE (3)** critical use cases (scenarios) related to security.

#### **3. Artifact Analysis (Step 3 and Step 4 – Perform Risk Assessment)**

- **Analyse existing documents** or systems to identify security risks.

#### **To Do:**

Any documentation could be used to uncover vulnerabilities?

#### **4. Risk Assessment (Step 4 - Perform Risk Assessment)**

- Identify and prioritize potential security risks.

**To Do:**

- a. Identify **THREE (3)** possible threats (e.g., phishing, man-in-the-middle attacks).
- b. **Prioritise** the threats based on their impacts.

#### **5. Questionnaires (Step 5 – Select Elicitation Techniques)**

- Develop questions to ask stakeholders about their security concerns.

**To Do:**

List **5 questions** to collect the security concerns from users of different levels.